Spacing–Capacity Correlation & Stop Popularity

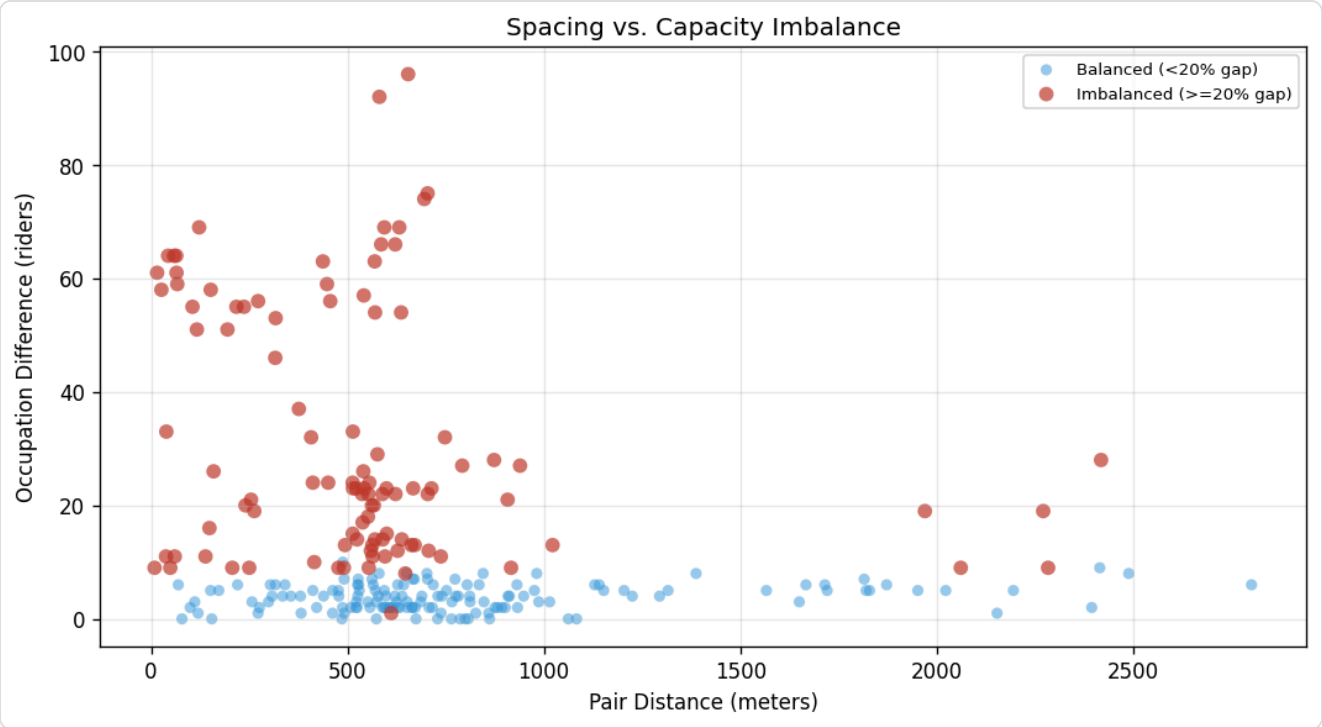
Data from 2026-03-16 14:50:19 to 2026-03-16 17:46:22 — 5,176 polls, 18 buses

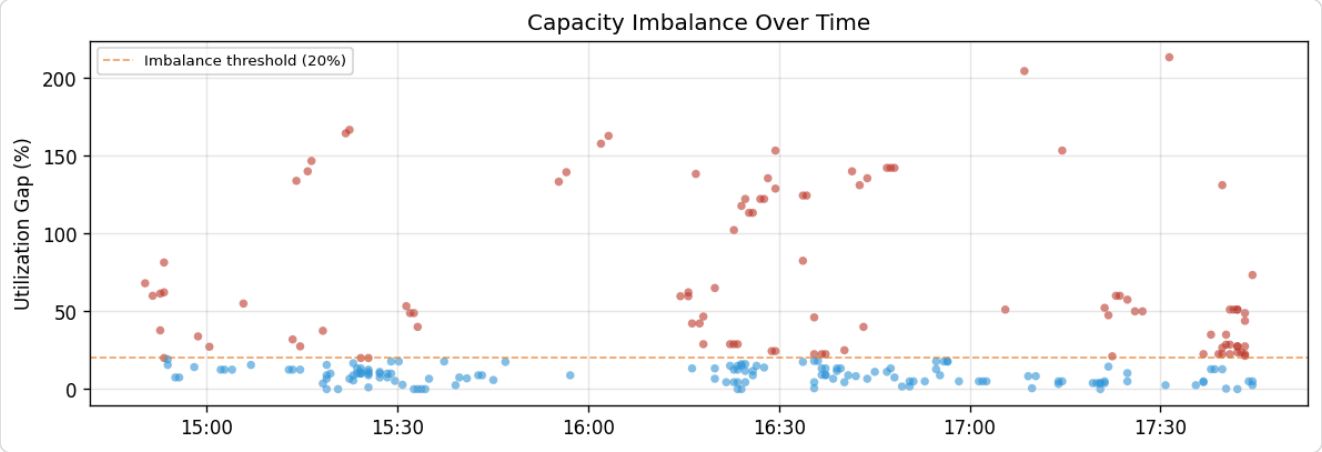
1. Spacing–Capacity Imbalance

At each spacing alert (same-direction, 25%+ below pair mean), we compare the two buses' ridership. A **capacity imbalance** is flagged when the utilization gap is ≥20% — one bus is significantly more loaded than the other, indicating bunching is hurting service distribution.

247	103	42%
Spacing Alerts w/ Capacity	Imbalanced (≥20% gap)	Imbalance Rate

213%
Worst Util Gap





Top Offending Pairs

Pair	Alerts	Imbalanced	Imbalance Rate	Worst Gap
2302 ↔ 2304	20	20	100%	153%
2304 ↔ 2307	11	9	82%	73%
2201 ↔ 2206	7	7	100%	60%
2302 ↔ 2307	7	7	100%	213%
2209 ↔ 2308	13	7	54%	68%
2308 ↔ 2309	12	7	58%	62%
2301 ↔ 2302	6	6	100%	163%
2211 ↔ 2308	7	6	86%	62%
2209 ↔ 2211	12	5	42%	28%
2203 ↔ 2208	4	4	100%	82%
2205 ↔ 2308	6	4	67%	81%
2204 ↔ 2211	9	4	44%	25%
2301 ↔ 2304	4	3	75%	34%
2202 ↔ 2206	3	2	67%	35%
2205 ↔ 2209	6	2	33%	23%
2209 ↔ 2309	11	2	18%	28%
2209 ↔ 2306	5	2	40%	32%

Pair	Alerts	Imbalanced	Imbalance Rate	Worst Gap
2211 ↔ 2309	10	2	20%	20%
2203 ↔ 2214	1	1	100%	65%
2301 ↔ 2307	2	1	50%	52%
2205 ↔ 2306	6	1	17%	46%
2211 ↔ 2306	4	1	25%	29%

Imbalance Events Log

▶ Show all 103 imbalance events

2. Low-Popularity Stop Discovery

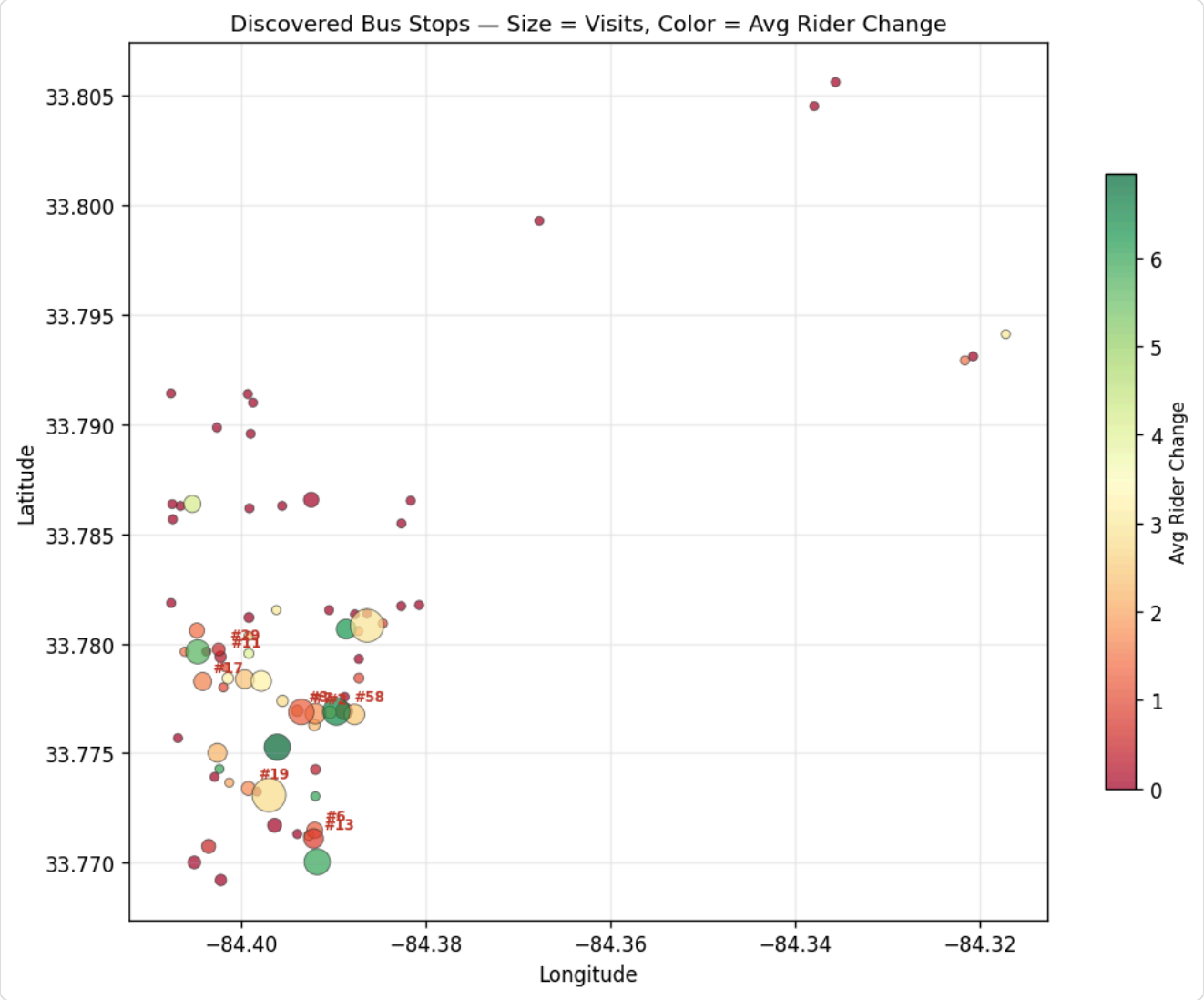
Auto-discovered stops from GPS data. A 'stop' requires speed=0 for 2+ consecutive polls (≥70s) to exclude red lights and crosswalks. Stops within 50m are clustered as the same physical stop. Rider change = |occupation after - occupation before| each visit.

78
Stops Discovered

431
Total Stop Events

10
Low-Popularity Stops

1.4
Avg Rider Change (all stops)



Flagged Low-Popularity Stops

Stops with avg rider change <2 AND visited by 3+ different buses. These are candidates for schedule optimization or removal.

Stop	Location	Visits	Unique Buses	Avg Rider Change	Avg Dwell	Popularity Score
#11	33.77939, -84.40226	4	3	0.2	16m 46s	1.0
#29	33.77974, -84.40245	5	4	0.5	1m 38s	2.5
#3	33.77694, -84.39395	4	3	1.2	1m 18s	5.0
#58	33.77692, -84.38887	9	4	0.9	1m 25s	8.0

Stop	Location	Visits	Unique Buses	Avg Rider Change	Avg Dwell	Popularity Score
#6	33.77149, -84.39206	8	4	1.2	1m 10s	10.0
#13	33.77111, -84.39218	12	5	0.8	1m 15s	10.0
#19	33.77339, -84.39925	6	5	1.7	1m 56s	10.0
#17	33.77827, -84.40419	10	3	1.6	3m 5s	16.0
#2	33.77680, -84.39199	13	9	1.8	1m 39s	23.0
#7	33.77688, -84.39351	20	9	1.2	1m 18s	25.0

All Discovered Stops (ranked by popularity, lowest first)

▶ Show all 78 stops